

Insurance Quote Funnel BI Analysis

QuoteFlow Insights — Insurance Quote Funnel BI Analysis

Role: Data Analyst

Tools: SQL-style aggregation, Google Sheets (SUMIFS/COUNTIFS/REGEXEXTRACT)

Dataset: Anonymized Commercial Insurance Quote Funnel (15,000 rows)

Project Background: An anonymized Canadian commercial-insurance MGA's online quoting funnel produced 15,000 applications spanning 13 industry categories and 9 coverage lines, moving from initial business details through a final underwriting decision. Leadership needed a structured read on three questions: how much revenue the book actually generates and from whom, why the funnel converts only a fraction of applicants, and where the digital customer journey itself is leaking volume. This report answers 90 business questions across those three areas, each tied to an explicit denominator, a computed result, and a recommended action.

Section 1: Premium Analysis

1. Total Priced Premium

Q: What is the Total Premium (Revenue) across all priced quotes?

Business Context: Leadership needed a current run-rate revenue baseline before evaluating growth levers.

Sheets Formula: =SUMIFS(TotalPremium, IsPriced, 1)

Result: \$4,179,161.11 across 2,087 priced quotes, only 13.9% of all 15,000 applications.

Action: use conversion improvement, not just new volume, as the primary growth lever, since 86% of applications never generate a dollar.

2. Average Priced Premium

Q: What is the Average Total Premium per priced quote?

Business Context: An average ticket size sets the ceiling for sustainable acquisition cost per customer.

Sheets Formula: =AVERAGEIFS(TotalPremium, IsPriced, 1)

Result: \$2,002.47. Action: cap cost-per-acquisition at a small fraction of this figure (e.g. under \$150-200) so unit economics hold before renewal lifetime value is even considered.

3. Median Priced Premium

Q: What is the Median Total Premium?

Business Context: Comparing mean to median reveals whether the book is skewed by a few large accounts.

Sheets Formula: =MEDIAN(FILTER(TotalPremium, IsPriced=1))

Result: \$1,677.94, about 19% below the \$2,002.47 mean, confirming a right-skewed book.

Action: budget and forecast off the median, but report both figures so the skew stays visible.

4. Premium Range

Q: What is the minimum and maximum priced premium?

Business Context: Extremes need to be checked for data errors before being treated as real accounts.

Sheets Formula: =MIN(FILTER(TotalPremium, IsPriced=1)) / =MAX(...)

Result: Min \$909.46, Max \$11,571.06 (6.9x the median). Neither looks like a data glitch. Action: flag the max-premium account for retention priority; check whether the minimum clears the cost-to-service bar.

5. Quotes With No Premium

Q: How many quotes have no premium at all, and why?

Business Context: "No premium" blends three very different situations that need separate owners.

Sheets Formula: =COUNTIFS(TotalPremium, "", ...) segmented by funnel-stage and rejection-log status

Result: 12,913 of 15,000 (86.1%): 5,012 abandoned mid-funnel, 2,457 explicitly rejected, and 5,444 "ghost complete" (finished every stage, no price, no logged reason). Action: prioritize the ghost-complete investigation first, it is the largest, most invisible bucket and likely a system gap rather than a UX or policy issue. Owner: Data/Engineering.

6. Premium Standard Deviation

Q: What is the standard deviation of priced premium?

Business Context: Spread determines whether one pricing/service model fits every customer.

Sheets Formula: =STDEV(FILTER(TotalPremium, IsPriced=1))

Result: \$1,102.30, about 55% of the mean. Action: test a segmented service model (streamlined for small accounts, hand-underwritten for larger ones) rather than one-size-fits-all.

7. Highest Total-Premium Category

Q: Which Category generates the highest total premium?

Business Context: Identifies which segment to protect first if acquisition budget is constrained.

Sheets Formula: =SUMIFS(TotalPremium, Category, "B&P Services", IsPriced, 1)

Result: B&P Services, \$1,091,485 (26% of revenue), then C.E.&I. (\$776,749) and Hospitality (\$754,946). B&P Services wins on volume (673 quotes), not price per policy. Action: protect the B&P Services funnel first since losing it hurts total revenue most, but don't assume it is the most profitable segment per account.

8. Highest Average-Premium Category

Q: Which Category has the highest average premium?

Business Context: Identifies where to focus high-value account acquisition.

Sheets Formula: =AVERAGEIFS(TotalPremium, Category, "C. E. & I.", IsPriced, 1)

Result: Farming leads at \$2,962 but on a very small sample; C.E.&I. (\$2,764) and Hospitality (\$2,451) are the real high-ticket segments with meaningful volume. Action: prioritize C.E.&I. and

Hospitality for high-value acquisition; don't build strategy around Farming until volume grows.

9. Lowest Average-Premium Category

Q: Which Category has the lowest average premium?

Business Context: Distinguishes a genuine low-ticket segment from a small-sample artifact.

Sheets Formula: =AVERAGEIFS(TotalPremium, Category, "Warehousing", IsPriced, 1)

Result: Warehousing (\$1,296, n=8, too small to trust) then B&P Services (\$1,622, n=673, reliable). Action: treat B&P Services as the real low-ticket segment; price and service it accordingly at scale.

10. Most Common Priced Category

Q: Count of priced quotes per Category, which is most common?

Business Context: Concentration in one category affects capacity planning and reinsurance sizing.

Sheets Formula: =COUNTIFS(Category, "B&P Services", IsPriced, 1)

Result: B&P Services leads (673 of 2,087, 32%), then Health Serv. (337) and Retail (315).

Action: size underwriting appetite and reinsurance treaties around B&P Services being the dominant segment.

11. Construction Segment Premium

Q: What is the total premium for Construction (Industry Code 15xx)?

Business Context: Tests whether a typically high-premium line is meaningfully represented in this book.

Sheets Formula: =SUMIFS(TotalPremium, IndustryCode, "15*", IsPriced, 1)

Result: \$68,170 across 23 priced quotes, 1.6% of total revenue. Action: treat Construction as a near-greenfield opportunity if leadership wants to grow it, otherwise deprioritize marketing spend there.

12. Hospitality vs. B&P Services Premium

Q: Compare Hospitality vs. B&P Services average premiums.

Business Context: Different segment economics justify different acquisition-cost ceilings.

Sheets Formula: =AVERAGEIFS(TotalPremium, Category, "Hospitality") vs Category="B&P Services"

Result: Hospitality \$2,451 vs B&P Services \$1,622, a 51% gap. Action: allow a higher acquisition-cost ceiling for Hospitality leads than for B&P Services leads.

13. Top-3-Category Revenue Share

Q: What % of total revenue comes from the Top 3 Categories?

Business Context: Quantifies category-level concentration risk.

Sheets Formula: =SUM of top 3 SUMIFS results / SUM(TotalPremium)

Result: 62.8% (\$2,623,180 of \$4,179,161). Action: concentrate underwriting expertise and marketing budget on the top 3 categories while continuing to monitor the remaining 10.

14. Premium vs. Revenue Band

Q: Does premium rise with the annual_revenue_band?

Business Context: Tests whether business size is a trustworthy pricing/lead-scoring signal.

Sheets Formula: =AVERAGEIFS(TotalPremium, RevenueBand, "<100K") repeated per band

Result: Yes, clearly monotonic: <100K \$1,315 → 100K-250K \$1,546 → 250K-500K \$1,813 → 500K-1M \$2,331 → 1M-5M \$3,107. Action: use revenue band directly in lead qualification and pricing models; it is the most reliable driver found in this book.

15. Liability (GL) Premium Share

Q: What is the total Liability (GL) premium (on-prem + products)?

Business Context: Identifies the single largest coverage line by dollars.

Sheets Formula: =SUMIFS(GLOnPrem)+SUMIFS(GLProducts)

Result: \$1,639,131, 39.2% of total premium. Action: prioritize GL rate review and reinsurance attention here first, since accuracy on this line has outsized revenue impact.

16. Flood Coverage Revenue Share

Q: What percentage of total revenue is Flood coverage?

Business Context: Tests whether Flood is a meaningful current revenue driver.

Sheets Formula: =SUMIFS(FloodPremium)/SUM(TotalPremium)

Result: \$33,288, 0.8% of total premium. Action: don't invest underwriting or product effort in Flood as a base-package driver today.

17. Zero-Flood Quotes

Q: How many priced quotes carry \$0 Flood premium?

Business Context: Quantifies the size of a potential cross-sell gap.

Sheets Formula: =COUNTIFS(FloodPremium, 0, IsPriced, 1)

Result: 1,516 of 2,087 (72.6%). Action: keep Flood a la carte, and evaluate a targeted flood cross-sell to the uncovered 72.6% rather than folding it into the base package.

18. Content vs. Building Premium Ratio

Q: What is the ratio of Content to Building premium?

Business Context: Indicates whether the customer base is mostly tenants or property owners.

Sheets Formula: =SUMIFS(ContentPremium)/SUMIFS(BuildingPremium)

Result: Content \$1,251,462 vs Building \$10,823, roughly 116:1. Action: treat building/property-owner coverage as a minor add-on, not a core product; prioritize tenant-focused development instead.

19. Highest Crime-Premium Category

Q: Which Category pays the most for Crime coverage?

Business Context: Checks whether crime premium tracks category volume or a distinct risk pattern.

Sheets Formula: =SUMIFS(CrimePremium, Category, "B&P Services")

Result: B&P Services (\$87,108), then C.E.&I. (\$53,740) and Hospitality (\$52,408) — the same top-3 categories by volume. Action: no crime-specific action needed beyond what already applies to the top 3 categories.

20. Top-Decile Premium Cutoff

Q: Top 10% of priced quotes: what is the premium cutoff?

Business Context: Defines a working threshold for "large account" treatment.

Sheets Formula: =PERCENTILE(FILTER(TotalPremium, IsPriced=1), 0.9)

Result: \$3,248. Action: use \$3,248+ as the working definition of a large account for relationship management, while expecting real variation within that tier.

21. Revenue Concentration (80% Threshold)

Q: What % of priced quotes contribute 80% of premium revenue?

Business Context: Tests whether the book is whale-dependent or broadly diversified.

Sheets Formula: Sort premium descending, running SUM until 80% of total is reached

Result: 66.1% of priced quotes (1,379 of 2,087) are needed to reach 80% of revenue, not a classic 80/20 book. Action: state this explicitly to leadership; diversification is a real strength that Q4/Q20 alone could wrongly obscure.

22. GL Rate Increase Scenario

Q: If we raised GL rates 5%, how much new revenue?

Business Context: Sizes a low-effort pricing lever on the largest coverage line.

Sheets Formula: =SUMIFS(GLTotal)*0.05

Result: ~\$81,957 (5% of \$1,639,131), about 2% lift to total book revenue. Action: model this against expected quote-volume elasticity before committing.

23. Coverage Mix by Category

Q: Build a coverage-mix table by Category (% of each category's premium by coverage).

Business Context: Tests whether coverage design is standardized or tailored per industry.

Sheets Formula: Pivot table, coverage components as % of row-category total premium

Result: Every category is dominated by the same core coverages (GL ~34-44%, business interruption ~15-19%, content ~30-40%), a standardized package rather than a tailored one. Action: consider whether category-specific risk justifies tailoring coverage mix and pricing by category.

24. Single-Coverage-Dominant Quotes

Q: How many quotes are "single-coverage-dominant" (one coverage > 70% of total)?

Business Context: Flags potential rating-engine edge cases.

Sheets Formula: =COUNTIFS(MAX(coverage components)/TotalPremium, ">0.7")

Result: 50 quotes (2.4% of priced): 27 GL-dominant, 20 content-dominant, 3 crime-dominant.

Action: sanity-check these 50 for correct rating rather than treating them as a priority segment.

25. Top-5-Account Stress Test

Q: If the top 5 priced accounts left, what % of revenue disappears?

Business Context: Directly tests account-level concentration risk.

Sheets Formula: =SUM(LARGE(TotalPremium, {1,2,3,4,5}))/SUM(TotalPremium)

Result: \$55,098, just 1.3% of total premium. Action: focus retention effort on segments (e.g., B&P Services volume) rather than any individual large account, confirming Q21's diversification finding.

26. Coverage Correlation Matrix

Q: Which coverages are bought together (9x9 correlation matrix)?

Business Context: Identifies natural product bundles already implicit in buying behavior.

Sheets Formula: =CORREL() across each coverage-component pair

Result: Strongest pairs: content & EBI (0.83), on-prem GL & business interruption (0.78). Building and Flood barely correlate with anything. Action: formally bundle content+EBI and GL+BI as named packages with bundle pricing to simplify the quote flow.

27. Property vs. Casualty Split

Q: What is the property vs. casualty split of the whole book?

Business Context: Informs reinsurance and capital-planning balance.

Sheets Formula: =SUM(property components)/SUM(TotalPremium) vs casualty components

Result: Property 53.2%, Casualty/Liability 39.2%, Crime 7.6%, a fairly balanced commercial book. Action: plan reinsurance and capital around a roughly even property/casualty split.

28. Average Coverages per Quote

Q: What is the average number of coverages per priced quote?

Business Context: Suggests a natural base-package plus add-on product structure.

Sheets Formula: =AVERAGE(COUNTIF(coverage components, ">0")) per row

Result: 6.12 of 9 possible coverages on average, implying 5 near-universal coverages plus 1 optional add-on. Action: formalize the 5 near-universal coverages as a default base package and present the rest as explicit optional add-ons.

29. Premium by Business Tenure

Q: How does average premium vary by years_in_business?

Business Context: Tests tenure as an alternate pricing signal to revenue band.

Sheets Formula: =AVERAGEIFS(TotalPremium, YearsInBusiness, "0-2") repeated per band

Result: No clean trend (0-2 yrs \$2,156, 3-5 yrs \$2,211, 6-10 yrs \$1,792, 11-20 yrs \$1,855, 20+ yrs \$1,942). Action: don't use tenure alone as a pricing signal; revenue band (Q14) is the more trustworthy driver.

30. Coverage Attach Rates

Q: What is the attach rate of each coverage?

Business Context: Confirms which coverages are effectively mandatory versus optional in practice.

Sheets Formula: =COUNTIF(coverage component, ">0")/COUNT(IsPriced=1)

Result: Crime, EBI, GL, business interruption all ~100%; content 99.7%; equipment 44.3%; products GL 29.0%; flood 27.4%; building 12.0%. Action: redesign the quote flow around a clear 5-coverage base package plus 4 named optional add-ons.

31. Priced-Quote Volume Trend

Q: Are priced-quote volumes trending up or down over time?

Business Context: Distinguishes a genuine trend from a data-cutoff artifact.

Sheets Formula: Pivot: COUNTIFS(IsPriced, 1) grouped by month

Result: Roughly flat, 202-244 priced quotes/month June-December 2025; January 2026's apparent drop is a partial-month artifact (data cuts off Jan 15). Action: no volume-trend action needed; flag January as incomplete in any leadership report.

32. Premium by Sales Channel

Q: Does sales channel (device_type / source_app) affect average premium?

Business Context: Tests whether channel quality shows up in ticket size, not just conversion.

Sheets Formula: =AVERAGEIFS(TotalPremium, SourceApp, "RDC") per channel/device

Result: Device type barely matters (within 1%). source_app shows real spread: one channel averages ~10-12% higher tickets than the others. Action: investigate why that channel produces higher-value quotes and consider shifting acquisition spend/terms toward it.

33. Premium as % of Revenue

Q: Premium as % of annual revenue (rate adequacy proxy, using band midpoints)?

Business Context: A rough proxy for whether larger accounts are underpriced relative to size.

Sheets Formula: =AVERAGEIFS(TotalPremium, RevenueBand, "<100K")/BandMidpoint, per band

Result: <100K band: 2.63% of revenue; drops steeply to 0.10% at 1M-5M, over 25x spread. Action: check whether the low ratio for large accounts reflects genuinely lower risk or under-pricing.

34. Most Valuable Industry Code

Q: Which single Industry Code is most valuable, and does it match its Category's behavior?

Business Context: Checks whether a Category label hides very different sub-businesses.

Sheets Formula: =SUMIFS(TotalPremium, IndustryCode, "5813")

Result: Industry Code 5813 (Hospitality), \$398,501 across 155 quotes, average \$2,571, closely matching Hospitality's category average (\$2,451). Action: target Industry Code 5813 for growth; it is both large and behaves consistently with its category.

35. Three Findings for the CEO (Premium)

Q: Putting it together: three premium findings for the CEO, each with a number and an action.

Business Context: Synthesizes the section into leadership-ready statements.

Sheets Formula: Synthesis of Q1-Q34

Result: (1) The funnel, not the market, is the bottleneck: 86% of applications never reach a priced quote. Action: fix the funnel before spending more on acquisition. (2) The book is diversified, not whale-dependent: it takes 66% of priced quotes to reach 80% of revenue, and losing the top 5 accounts costs only 1.3%. Action: focus retention on segments, not individual accounts. (3) Revenue band, not tenure or category, is the reliable price driver, rising cleanly from \$1,315 to \$3,107 across bands. Action: use revenue band as the primary lead-scoring signal.

Section 2: Rejection & Data Quality

36. Overall Rejection Rate

Q: What is the overall Rejection Rate, and what counts as a "real" completed quote?

Business Context: The "real" rate depends entirely on the chosen denominator, and different audiences need different framings.

Sheets Formula: =COUNTIFS(RejectionLog,"<>") /
(COUNTIFS(RejectionLog,"<>")+COUNTIFS(IsPriced,1))

Result: 54.2% of quotes that reached a real final outcome (2,461 rejected of 4,544 = 2,461 rejected + 2,087 priced). Across all 15,000 applications, rejected quotes are only 16.4%. Action: always report rejection rate with its denominator stated explicitly, use 54.2% for underwriting-appetite conversations and 16.4% for funnel/growth conversations. Owner: Analytics.

37. Rejected / Priced / Ghost-Completed Counts

Q: How many quotes were truly rejected, priced, or ghost-completed?

Business Context: Blending these three outcomes misdirects the fix toward the wrong team.

Sheets Formula: =COUNTIFS(RejectionLog,"<>[]","<>") / =COUNTIFS(IsPriced,1) /
=COUNTIFS both blank

Result: Rejected 2,461, Priced 2,087, Ghost-completed 5,444 (finished every stage, no premium, no reason) — plus 5,012 real funnel dropouts account for all 15,000 rows. Action: assign three separate owners (Underwriting for rejected, Data/Engineering for ghost-complete, nothing needed for priced) and never blend ghost-complete into rejected.

38. Category Rejection Rate Reliability

Q: Which Category has the highest rejection rate, and which category rates can you actually trust?

Business Context: A 100% rate on 2 quotes is noise; the same rate on 650 quotes is a real signal.

Sheets Formula:

=COUNTIFS(Category,"Realty",Rejected,1)/COUNTIFS(Category,"Realty",Resolved,1), filtered to n≥30 resolved

Result: Nine categories show 100% rejection among decided quotes, but only some (Transport n=117, Utilities n=65, Mfg./Proc. n=80, Recreation n=68, Farming n=38, Warehousing n=30) have enough volume to trust. Realty is the standout at 99.1% on n=654, high enough volume to

be a real "we don't write this" finding. Action: stop marketing spend on Realty leads immediately. Owner: Underwriting + Marketing.

39. Industry Code Cross-Field Match

Q: Does the applicant's Industry Code match the `ibc_code` in its rejection reason?

Business Context: A high match rate makes the derived log field trustworthy for downstream use.

Sheets Formula: `=COUNTIFS(IndustryCode, IBCFromLog)/COUNTIFS(both populated)`

Result: 98.0% match rate (1,885 of 1,923). Action: use `ibc_from_log` as an automated data-quality tripwire, flagging any future mismatch for manual review.

40. Multi-Reason Rejections

Q: How many rejections cite multiple reasons, and what's the distribution?

Business Context: Bounds how many rejections could realistically convert if a single rule were relaxed.

Sheets Formula: `=REGEXEXTRACT`-based distinct reason-type count per row, then `COUNTIFS` by count

Result: 1,607 rejections (65.3%) cite exactly one distinct reason; 854 (34.7%) cite two or more. Action: size any single-rule relaxation proposal against the 1,607 single-reason population only, since multi-reason declines won't convert from one rule change.

41. FUS Grade Distribution

Q: What is the distribution of FUS (fire protection) grades among rejections?

Business Context: Tests whether fire-protection declines are a graduated scale or a hard cutoff.

Sheets Formula: `=REGEXEXTRACT(RejectionLog,"fus_grade": "(\\d+)")`, pivoted by grade

Result: Grade 8 = 13, Grade 9 = 271, Grade 10 = 320, of 604 total. No grades below 8 appear.

Action: confirm whether this is a documented hard cutoff at grade 8 or an emergent pattern, and formalize it as policy either way. Owner: Underwriting.

42. Grade-10 Rejections

Q: How many quotes were rejected with FUS Grade 10 (worst)?

Business Context: Isolates the single most severe fire-protection tier.

Sheets Formula: `=COUNTIFS(FUSGrade, 10)`

Result: 320 of 604 FUS-grade rejections (53.0%). Action: keep Grade 10 a flat decline; reserve conditional-acceptance testing for grades 8-9.

43. Most Common Decline Rule

Q: Which decline rule (`decline_uw_rule`) triggers most often?

Business Context: The most-triggered rule deserves the most underwriting scrutiny.

Sheets Formula: `=COUNTIFS(DeclineUWRule, 1.0)`

Result: Rule 1.0 fires most, 271 of 492 rule-tagged rejections (55.1%). Action: start any

decline-rule audit with Rule 1.0.

44. Top Decline Rule Profile

Q: What does the top decline rule target? Profile the rows it rejects.

Business Context: Determines whether Rule 1.0 is a fire-risk rule or an alert-driven rule.

Sheets Formula: Pivot: COUNTIFS(DeclineUWRule,1.0) crossed with Category and alert presence

Result: 100% of Rule 1.0 rejections carry an alert, only 26% also carry a FUS grade, so it is alert-driven, not fire-risk. It disproportionately hits Hospitality (27.3% of Rule 1.0 rejections vs 11.7% of rejections overall). Action: build a conditional-acceptance path for common hospitality hazards instead of a flat Rule 1.0 decline. Owner: Underwriting + Product.

45. Most Common Alert Types

Q: What are the most common underwriter_alert types, and what risks do they flag?

Business Context: Groups the alert taxonomy into actionable hazard families.

Sheets Formula: =REGEXEXTRACT(RejectionLog, alert-list pattern), deduplicated and pivoted

Result: 602 deduplicated alert occurrences across 14 types; top 5 are cranes/heavy equipment (109), hazardous materials (94), swimming pool (72), hibachi (66), flaming shots (66). Action: build two named underwriting tracks, Physical/Contractor Hazard and Hospitality Operational Hazard, since each needs different mitigation products.

46. Contractor-Alert Category Attribution

Q: Which Categories trigger "contractor_alerts", and does every case make obvious sense?

Business Context: An unexpected category triggering a hazard alert may signal a data-entry error rather than genuine risk.

Sheets Formula: =COUNTIFS(AlertList,"*contractor*",Category,"C. E. & I.")

Result: 173 rejections carry a contractor alert; C.E.&I. leads (39.9%), then Realty (30) and Hospitality (25). Hospitality/Member Org./Recreation carrying contractor alerts is less expected. Action: manually review the 39 out-of-pattern rows for correct categorization. Owner: Underwriting QA.

47. Underwriter Alerts and Acceptance

Q: Do underwriter alerts always lead to rejection, or can an accepted quote carry one?

Business Context: Tests whether this is a business insight or a data-structure artifact.

Sheets Formula: =COUNTIFS(Status,"accepted",AlertList,"<>")

Result: Always, structurally: the alert field only ever exists inside a non-empty rejection log, so this is tautological, not an independent finding. Action: don't present this as an underwriting insight; focus on which alert types are most severe instead.

48. Eligibility vs. Hazard Rejections

Q: How many rejections are driven by size/eligibility reasons rather than hazard?

Business Context: Eligibility declines are a policy lever; hazard declines are a risk judgment.

Sheets Formula: =REGEXMATCH(RejectionLog,"revenue|content_limit|provincial_restriction")

Result: Of 2,461 rejections: 149 (6.1%) eligibility, 1,015 (41.2%) hazard, 25 both, 1,322 (53.7%) neither (ibc_code or a bare rule number only). Action: prioritize the 149 eligibility declines as the cheapest, most defensible fix; investigate the unexplained 1,322 before writing them off.

49. Google Place ID Pricing Gate

Q: Is a Google Place ID required to get priced? Test whether verification is a hard gate.

Business Context: Determines whether verification is the funnel's real bottleneck.

Sheets Formula: =COUNTIFS(IsPriced,1,PlaceID,"")

Result: Yes, 100% of priced quotes have a Place ID, zero exceptions. But 58.4% of unpriced quotes already have one too, so verification is necessary but far from sufficient. Action: don't invest further in Place-ID capture as a pricing-volume lever; focus on stages after verification instead. Owner: Product/Growth.

50. Verified vs. Unverified Rejections

Q: How many rejected quotes are verified (have a Place ID) vs. unverified?

Business Context: Confirms whether rejection is a verification problem or an underwriting-appetite problem.

Sheets Formula: =COUNTIFS(Rejected,1,PlaceID,"<>")

Result: 2,338 of 2,461 (95.0%) are verified. Action: treat verification (Data/Product) and acceptance rate (Underwriting) as independent problems with independent owners.

51. Rejection Rate by Source Channel

Q: Does rejection rate differ by source platform (source_app)?

Business Context: Distinguishes channels that are both high-volume and high-risk from genuinely good partners.

Sheets Formula:

=COUNTIFS(SourceApp,"RDC",Rejected,1)/COUNTIFS(SourceApp,"RDC",Resolved,1)

Result: Best channel rejects least (43.3%), a second channel highest among tracked channels (57.0%) while also bringing the most volume, the worst combination for acquisition spend.

Action: reallocate partner spend toward the low-rejection channel and renegotiate the high-volume, high-rejection one. Owner: Partnerships/Marketing.

52. Lowest-Rejection Category

Q: Which Category has the lowest rejection rate, our easiest business to write?

Business Context: Identifies the best combination of low risk and reliable volume.

Sheets Formula: =COUNTIFS(Category,"Health Serv.",Rejected,1)/COUNTIFS(Category,"Health Serv.",Resolved,1), filtered to n≥30

Result: Health Serv. lowest at 12.0%, then B&P Services at 32.0%. Action: point acquisition spend at Health Serv. first; use B&P Services as the volume backstop.

53. Rejection Rate Over Time

Q: Is the rejection rate changing over time (created_at)?

Business Context: Distinguishes a real trend from noise before recommending an intervention.

Sheets Formula: Pivot: rejection rate by month(CreatedAt)

Result: Drifts from 56.4% in June 2025 to 51.7% in December, a mild ~5-point improvement over 8 months. Action: instrument a monthly rejection-rate dashboard; revisit only if the slope steepens. Owner: Analytics.

54. Gross Lost Premium from Rejections

Q: Estimate the lost premium from rejected quotes.

Business Context: A gross figure is dramatic but misleading if presented as recoverable.

Sheets Formula: =COUNTIFS(Rejected,1)*AVERAGEIFS(TotalPremium,IsPriced,1)

Result: ~\$4.93M gross (2,461 x \$2,002.47 average premium), most of it correctly declined risk. Action: always pair this with the salvageable single-reason subset (~\$2.65M, Q56) rather than presenting the gross figure alone.

55. Single-Reason Rejection Recovery Scenario

Q: If we relaxed the single most common decline reason, how many quotes could we recover?

Business Context: Sizes the highest-leverage single policy change available.

Sheets Formula: =COUNTIFS(RejectionLog,"*ibc_code*") filtered to single-reason-only rows

Result: ibc_code is the most common reason (2,020 of 2,461 rejections); restricted to single-reason-only, 1,322 quotes worth ~\$2.65M. Action: pilot relaxing ibc_code eligibility for the most-declined industry codes and monitor loss ratio before making it permanent. Owner: Underwriting + Actuarial.

56. Edge Cases: Priced-and-Rejected, Reasonless Declines

Q: Find rejected quotes that carry a premium, or a decline with no reason.

Business Context: Confirms the dataset's audit trail is clean before trusting it further.

Sheets Formula: =COUNTIFS(Rejected,1,TotalPremium,">0") and
=COUNTIFS(Rejected,1,RejectionLog,"")

Result: 4 rows are both rejected and carry a premium (\$4,884.61 total); 0 rows are rejected with no reason logged. Action: add a distinct status flag for the 4 priced-then-declined rows and confirm intent with Underwriting.

57. Status-Column Audit

Q: Audit the status column: how many distinct values, and which are really the same thing?

Business Context: Messy categorical values are a sign of inconsistent data entry across systems.

Sheets Formula: =COUNTA(UNIQUE(Status))

Result: 9 raw values, only 6 real business states once casing/wording variants are collapsed.

Action: freeze an official 6-value status list with input validation at the source. Owner: Data/Engineering + Operations.

58. Status Cleaning Map

Q: Build a cleaning map for status: how many rows land in each clean bucket?

Business Context: Converts a messy categorical field into a trustworthy standard set.

Sheets Formula:

=COUNTIFS(Status,"referred")+COUNTIFS(Status,"quoted")+COUNTIFS(Status,"quoting")...

Result: Referred 5,791, Incomplete 4,881, Accepted 1,916, Quoted 1,361, Lost 979, Issued 72, reconciling exactly to 15,000. Action: store this raw-to-clean map as a maintained pipeline lookup, not a one-off spreadsheet formula. Owner: Analytics Engineering.

59. Reconciling Status, Rejection, and Incomplete Stage

Q: Reconcile the three outcome fields, status, rejection, and incomplete_stage. Do they agree?

Business Context: Determines which field is the single source of truth for a decline decision.

Sheets Formula: Pivot: Status crossed with Rejection flag

Result: They describe different moments, not the same fact. "Accepted" status splits across rejection=0, rejection=1, and rejection blank. Action: declare rejection as the single official decline field and status as workflow-only, published in the data dictionary. Owner: Analytics.

60. Data-Type Fixes

Q: Fix the data types: rejection and premium columns are stored as text. What breaks, and how do you coerce them?

Business Context: Text-typed numeric columns silently break SUM/AVERAGEIF with no error.

Sheets Formula: =VALUE(cell) wrapped coercion, or =ARRAYFORMULA(N(range))

Result: Numeric-looking strings mixed with true blanks silently understate SUM/AVERAGEIF totals in Sheets/Excel. Action: enforce a typed-schema step at ingestion so numbers-as-text never reach an analyst's spreadsheet. Owner: Data Engineering.

61. Three Kinds of Missing

Q: Distinguish the three kinds of "missing" in this data, blank, the literal "Missing", and a placeholder token. Why does it matter?

Business Context: Treating a typed placeholder the same as a true blank hides that a record was actually reviewed.

Sheets Formula: =COUNTIF(nfp_notes,"Missing") vs =COUNTBLANK(nfp_notes) vs remainder

Result: 11,298 rows contain the literal word "Missing", 1,594 are truly blank, 2,108 contain a real note, reconciling to 15,000. Action: adopt one explicit missing-data convention and report a completeness percentage per key column. Owner: Data Engineering + Analytics.

62. Normalizing "New Venture" Notes

Q: Normalize the free-text nfp_notes: how many quotes are really "New Venture" once casing and formatting are unified?

Business Context: An exact-match query can radically undercount a free-text-tagged segment.

Sheets Formula: =COUNTIF(nfp_notes,"New Venture") vs =COUNTIF(nfp_notes,"*new venture*")

Result: Exact match: 12 rows. Case-insensitive contains-match: 788 rows, a 65x difference. Action: replace the free-text note with structured Y/N flags at capture; ship a regex-based flag as an interim fix. Owner: Product + Analytics.

63. Duplicate Hunt

Q: Hunt for duplicates: are there exact duplicate rows, and how do repeat-email rows differ from true duplicates?

Business Context: A naive "remove duplicates" pass would delete a real, valuable customer segment.

Sheets Formula: =COUNTIFS(all columns match) for exact dupes; =COUNTIF(insured_email) > 1 for repeats

Result: 0 exact duplicate rows; 250 unique businesses applied more than once (92 twice, 73 three times, 85 four times), 493 extra rows. Action: never blanket-dedupe on email; dedup only for unique-customer counts, never for volume counts. Owner: Analytics.

64. IQR Outlier Detection

Q: Detect premium outliers the statistical way (IQR), and decide which to keep, cap, or investigate.

Business Context: Quantifies how much a handful of large accounts distort the headline average.

Sheets Formula: =QUARTILE(TotalPremium,1) / =QUARTILE(TotalPremium,3), fence = $Q3 + 1.5 * (Q3 - Q1)$

Result: Q1 \$1,250.00, Q3 \$2,408.24, upper fence \$4,145.60. 79 quotes (3.8%) exceed it; excluding them moves the mean from \$2,002.47 to \$1,858.22, a ~7% shift. Action: keep the outliers as real customers, but report the median and a trimmed mean alongside the raw mean; route any premium failing a sanity check to Underwriting. Owner: Analytics + Underwriting.

Section 3: Customer Journey & Digital Analytics

65. Full Funnel Stage Count

Q: What does the funnel look like? Count applicants at each stage from start to "complete quote".

Business Context: Identifies where volume is actually being lost across the nine-stage flow.

Sheets Formula: =COUNTIF(incomplete_stage, "stop at stage 1*") repeated per stage

Result: 9,988 of 15,000 reach the raw "complete quote" flag; the two biggest early cliffs are stage 1 (business location, 1,411) and stage 2 (annual revenue, 1,278), plus stage 8 (enter information, 1,122). Action: fix the top 1-2 leak stages this quarter rather than spreading effort across all nine. Owner: Product/UX.

66. True Completion Rate

Q: What is the TRUE completion rate, and why does the "complete quote" flag overstate it?

Business Context: The raw flag counts a form submission, not a real outcome.

Sheets Formula: =COUNTIFS(RejectionLog,"<>",TotalPremium,">0")/COUNT(all rows)

Result: Raw flag: 66.6%. True resolved-completion (rejection log OR premium > 0): 30.3%, a 2x overstatement. Action: adopt resolved-completion as the official funnel KPI and treat the 5,444 ghost-completes as a priority data-integrity defect. Owner: Analytics + Engineering.

67. Biggest Single Drop-Off Stage

Q: Which single stage has the highest drop-off, the biggest cliff?

Business Context: Identifies the single highest-leverage screen to redesign.

Sheets Formula: =MAX of per-stage COUNTIF(incomplete_stage)

Result: Stage 1, business location: 1,411 abandoners (9.4% of all applicants), the largest raw count of any stop-point. Action: redesign and A/B test the stage-1 screen. Owner: Product/UX.

68. Annual Revenue Stage Drop-Off

Q: What % of applicants drop off at the "annual revenue" stage (step 2)?

Business Context: A sensitive question asked too early can cost otherwise-good leads.

Sheets Formula: =COUNTIF(incomplete_stage,"stop at stage 2")/COUNT(all rows)

Result: 8.5% (1,278 of 15,000). Action: resequence the revenue question later in the flow, or soften it to a band-picker, and A/B test against the current rate. Owner: Product.

69. Liability Coverage Stage Drop-Off

Q: How many applicants stop at "liability coverage" (step 7)?

Business Context: A late-stage abandoner is a more expensive loss than an early one.

Sheets Formula: =COUNTIF(incomplete_stage,"stop at stage 7")

Result: 115 (0.8% of all applicants), the smallest raw drop-off but a highly engaged group.

Action: add plain-language coverage explanations and trigger a targeted follow-up email to these near-wins. Owner: Product/UX + Marketing.

70. Device-Type Drop-Off Comparison

Q: Do mobile users drop off differently than desktop users?

Business Context: Rules out (or confirms) a device/UX theory for the funnel's completion gap.

Sheets Formula: =AVERAGEIFS(ResolvedFlag, DeviceType, "desktop") per device

Result: No, resolved-completion is nearly identical: desktop 30.5%, mobile 30.1%, tablet 30.0%. Action: don't spend UX budget on device-specific redesigns; redirect that effort to the content fixes at stages 1 and 8.

71. Funnel Comparison by Source Platform

Q: Compare the funnel across source platforms (source_app).

Business Context: Identifies which acquisition channels bring genuinely well-qualified traffic.

Sheets Formula: =AVERAGEIFS(ResolvedFlag, SourceApp, "SN")

Result: Best channel resolves at 70.9%, nearly double the other two tracked channels (38.6% and 37.1%). The 32%-of-all-rows blank-source group resolves at just 2.0%. Action: invest in growing the top channel and renegotiate or re-integrate the underperformers. Owner: Partnerships.

72. Completion by Premium Tier

Q: Compare completion: high-premium sectors (Construction, Hospitality) vs. low (B&P

Services).

Business Context: Tests whether high-value business is also easy to close, or a harder sell.

Sheets Formula: =AVERAGEIFS(ResolvedFlag, Category, "C. E. & I.")

Result: C.E.&I. resolves at 62.7%, Hospitality at 51.5%, both higher than B&P Services (49.6%), the golden-segment scenario: high value and high conversion together. Action: feed the two golden segments more acquisition traffic rather than building a separate low-completion rescue flow. Owner: Marketing.

73. Google Place ID and Journey Outcome

Q: Where do applicants who lack a Google Place ID drop off or fail?

Business Context: Distinguishes a UX abandonment problem from a hidden system gate.

Sheets Formula: =AVERAGEIFS(ResolvedFlag, PlaceID, "") and
=AVERAGEIFS(FormCompleteFlag, PlaceID, "")

Result: They still complete the form 94.9% of the time, but resolve at only 2.3% and get priced at exactly 0.0%. Action: offer a manual-verification fallback for otherwise-legitimate applicants without a Place ID. Owner: Product + Engineering.

74. nfp_notes Category Resolve Rates

Q: What do the nfp_notes tell us (New Venture, Duplicate, Attempted Contact)?

Business Context: Free-text notes can hide a systemic bug or an underserved segment.

Sheets Formula: =AVERAGEIFS(ResolvedFlag, NoteCategory, "Duplicate") per category

Result: Duplicate-flagged applications resolve at just 15.6% (n=45), far below every other category, including New Venture (31.9%, n=788, the largest volume-of-concern bucket) and the ~45% overall baseline for other real notes. Action: send Duplicate cases to Engineering as a submission/tracking bug, and evaluate a startup-friendly manual-rate path for New Venture applicants. Owner: Engineering + Product/Underwriting.

75. Repeat Applicant Count

Q: How many businesses applied more than once (duplicate emails)?

Business Context: Identifies the size of a motivated-repeat-applicant segment.

Sheets Formula: =COUNTIF(insured_email) grouped, filtered to count > 1

Result: 250 unique businesses applied 2-4 times each. Action: capture email on the first attempt and build a nurture/save-and-resume flow targeting this segment specifically. Owner: Marketing + Product.

76. Repeat-Applicant Consistency

Q: For repeat applicants, is the business information consistent across their attempts?

Business Context: Confirms these are genuine repeat attempts, not shared-email fraud.

Sheets Formula: =COUNTIFS(email-grouped IndustryCode, Category, Province all match)

Result: All 250 repeat businesses (100%) show consistent Industry Code, Category, and province across attempts. Action: build a linking key so a returning applicant's history persists between attempts. Owner: Engineering.

77. Repeat-Applicant Eventual Outcome

Q: Did repeat applicants eventually complete, does persistence pay off?

Business Context: Tests whether the fix is "make it easier" or "encourage more attempts."

Sheets Formula: =COUNTIFS(email-grouped, EVER Resolved=1)/COUNT(unique repeat businesses)

Result: All 250 repeat businesses (100%) eventually reached "complete quote" and eventually resolved. Action: since persistence already works, ship save-and-resume so motivated customers finish in one visit instead of two to four. Owner: Product.

78. Completion Trend Over Time

Q: Are completions trending up or down over time (created_at)?

Business Context: Avoids over-claiming a trend from noisy monthly data.

Sheets Formula: Pivot: raw completion rate by month(CreatedAt)

Result: Ranges 63.8%-71.5% across full months with no sustained slope. Action: don't assert a trend; instrument monthly cohort tracking so future fixes are measurable against a clean baseline. Owner: Analytics.

79. Highest-Value Drop-Off Stage

Q: Which stage's drop-off costs the most in potential revenue, not just count?

Business Context: Tests whether the headcount-ranked priority also holds when weighted by dollars.

Sheets Formula: =COUNTIF(incomplete_stage,"stage 1")*AVERAGEIFS(TotalPremium,IsPriced,1)

Result: Stage 1 costs ~\$2.83M in potential premium, stage 2 ~\$2.56M, stage 8 ~\$2.25M, the same ranking as the headcount view. Action: fix stage 1 first, there is no revenue-vs-volume tradeoff to weigh here. Owner: Product/UX.

80. Province and Channel Efficiency

Q: Which province and which channel are most "efficient" (highest completion)?

Business Context: Determines whether a geographic reallocation is statistically defensible.

Sheets Formula: =AVERAGEIFS(ResolvedFlag, Province, "Ontario")

Result: Only Ontario has a meaningful sample (90.3% of rows); other provinces are too thin to rank reliably. By channel, the top channel is clearly most efficient at 70.9% resolved. Action: don't reallocate budget by province from this data; do reallocate toward the top-performing channel. Owner: Marketing.

81. Potential Revenue in the Incomplete Funnel

Q: Estimate the total potential revenue sitting in the incomplete funnel.

Business Context: A defensible recovery figure is more credible to leadership than a gross, unfiltered one.

Sheets Formula: =COUNTIFS(late-stage abandoners OR ghost-complete)*AVERAGEIFS(TotalPremium,IsPriced,1)

Result: Gross unresolved population is worth ~\$20.9M, but the defensible high-intent subset (late-stage abandoners plus ghost-completes, 6,941 applicants) is worth ~\$13.9M. Action: present the \$13.9M defensible figure, not the \$20.9M gross figure, as the business case for UX and ghost-complete fixes. Owner: Product + Finance.

82. Step-to-Step Micro-Conversion Rates

Q: Compute the step-to-step conversion rate between every consecutive stage, where is the weakest micro-conversion?

Business Context: Raw stage counts can hide a weak link buried in the middle of the funnel.

Sheets Formula: =COUNTIFS(reached stage N+1)/COUNTIFS(reached stage N) per consecutive pair

Result: Step 1→2 continue-rate 90.6%; steps 3-7 all hold 96.9%-99.0%; step 8→9 drops to 90.1%, the second-weakest transition in the funnel. Action: prioritize fixing the stage 8→9 transition alongside stage 1, and stand up a permanent step-conversion dashboard. Owner: Product + Analytics.

83. finished_percentage as a Progress Metric

Q: Use finished_percentage as a progress metric: what's the distribution, and does it agree with incomplete_stage?

Business Context: Clarifies whether the metric is trustworthy for external reporting or lead scoring.

Sheets Formula: =AVERAGE(finished_percentage), =MEDIAN(finished_percentage), grouped by stage

Result: Mean 63.5%, median 80% (the metric caps at 80, never reaching 100), rising monotonically with stage as expected. Action: use finished_percentage to score how "warm" an abandoner is for remarketing, and confirm with Product why the metric caps at 80. Owner: Marketing + Product.

84. Stage-1 Bounce Rate

Q: Calculate the bounce rate: what share of applicants leave at the very first step without engaging?

Business Context: A stage-1 leaver never really engaged and needs a different owner than a mid-funnel abandoner.

Sheets Formula: =COUNTIF(incomplete_stage,"stop at stage 1*")/COUNT(all rows)

Result: 9.4% (1,411 of 15,000). Action: route bounce (a Marketing/traffic-quality problem) and mid-funnel drop-off (a Product/UX problem) to different teams. Owner: Marketing vs. Product.

85. Channel-Attribution Gap Investigation

Q: Investigate the channel-attribution gap: ~32% of rows have a blank source_app. What are those rows really?

Business Context: A metric that looks like a top performer can actually be the ghost-data population in disguise.

Sheets Formula: =COUNTIFS(SourceApp,"",FormCompleteFlag,1) /
=COUNTIFS(SourceApp,"",GhostComplete,1)

Result: 4,793 rows (32.0%) have a blank source; 99.6% look "complete" but only 2.0% actually resolve, and 97.6% of them are ghost-completes. Action: report channel KPIs on tracked, resolved rows only, and treat this group as the primary evidence for the ghost-complete engineering investigation. Owner: Engineering.

86. Device x Channel Interaction

Q: Build a device x channel segmentation grid for completion, is there an interaction effect?

Business Context: Tests whether a low-performing channel is actually a device-specific integration problem.

Sheets Formula: Pivot: AVERAGEIFS(ResolvedFlag) crossed by DeviceType and SourceApp

Result: The top channel runs 69.6%-72.0% resolved on every device; the weaker channels run consistently lower regardless of device. Action: invest in improving or renegotiating the low-completing partner integrations rather than any device-specific UX work. Owner: Partnerships.

87. Resolved Quotes: Priced vs. Rejected Split

Q: Of quotes that actually resolved, what share got a price vs. were rejected?

Business Context: Cross-verifies the rejection rate from the journey side of the funnel.

Sheets Formula: =COUNTIFS(IsPriced,1)/COUNTIFS(Resolved,1)

Result: Of 4,544 resolved quotes, 45.9% (2,087) priced and 54.2% rejected (2,461), the exact mirror of the Section 2 rejection-rate finding once ghost-completes are excluded. Action: grow revenue by shifting salvageable rejections (single-reason, eligibility-only) into priced outcomes, since those customers already finished the funnel. Owner: Underwriting.

88. Monthly Cohort Completion Trend

Q: Cohort the applicants by created-month and compare their completion, is the funnel improving over time?

Business Context: Checks the raw-completion trend against the same honesty bar as Q78.

Sheets Formula: Pivot: raw completion rate by month(CreatedAt)

Result: Same 63.8%-71.5% range as Q78, no strong sustained slope. Action: don't claim a trend; stand up ongoing monthly cohort tracking. Owner: Analytics.

89. Started-to-Priced Conversion Rate

Q: Compute the overall started→priced conversion rate, the true top-to-bottom funnel efficiency.

Business Context: The single north-star number that summarizes the whole funnel in one figure.

Sheets Formula: =COUNTIFS(IsPriced,1)/COUNT(all rows)

Result: 13.9% (2,087 of 15,000), which decomposes as 30.3% resolve x 45.9% of resolvers get priced. The completion hop (30.3%) is the larger of the two losses. Action: adopt started→priced (~14%) as the north-star funnel KPI and prioritize fixing the completion hop over the resolved-to-priced split. Owner: Product + Analytics.

90. Three Journey Findings for the CEO

Q: Putting it together: top 3 journey findings for the CEO, each with a number and a fix.

Business Context: Synthesizes the customer-journey section into leadership-ready statements.

Sheets Formula: Synthesis of Q65-Q89

Result: (1) Only 30.3% of applicants ever reach a real outcome, 5,444 "complete" quotes are silent system failures worth an estimated \$13.9M in defensible lost premium. Fix: treat as the #1 engineering priority. (2) The biggest single leak is stage 1 (business location), losing 1,411 applicants worth ~\$2.83M, and it's one of only two weak points in the entire step-by-step funnel. Fix: redesign and A/B test the opening screen. (3) Channel quality, not device, drives outcomes: the top channel converts at 70.9% versus 37-39% for the others, and is simultaneously the lowest-rejection channel. Fix: shift acquisition spend toward it and renegotiate the underperformers. Owner: cross-functional (Engineering, Product, Partnerships), sponsored by Analytics for measurement.